

PLATFORM INFRASTRUCTURE COST ANALYSIS (0 TO 1,000,000 STUDENTS)

Comparative Cost Matrix: AI Engine Models vs. TiDB Cloud Serverless Database | Strictly Two Core Tables

CLARIFICATION ON AI COST (\$1,030 vs. <\$5.00): The **\$1,030/mo** figure represented a theoretical worst-case where 5% of 10,000,000 quizzes (500,000 live API calls) bypass cache and hit live Gemini every month. The **<\$5.00/mo** figure represents our actual TiDB Question Bank architecture: the Egyptian curriculum is finite (~300 chapters). Once questions are generated into TiDB, 1,000,000 students take quizzes directly from the database without invoking live Gemini API calls.

TABLE 1: AI ENGINE MONTHLY COST SCALE (OUR CURRENT PLAN vs. OTHER AI MODELS)

Scale (Active Students)	Monthly Quizzes (10 / Student)	OUR CURRENT PLAN Gemini 1.5 Flash + DB Cache	Gemini 1.5 Flash 100% Live (No Cache)	OpenAI GPT-4o mini 100% Live AI	Gemini 1.5 Pro 100% Live AI	OpenAI GPT-4o (Full) 100% Live AI	Claude 3.5 Sonnet 100% Live AI
0 – 1,000 (Launch)	10,000	\$0.00 / mo (Free Tier)	\$25.00	\$51.00	\$425.00	\$850.00	\$1,150.00
10,000 Students	100,000	\$0.00 / mo (Served from DB)	\$250.00	\$510.00	\$4,250.00	\$8,500.00	\$11,500.00
50,000 Students	500,000	< \$1.00 / mo	\$1,250.00	\$2,550.00	\$21,250.00	\$42,500.00	\$57,500.00
100,000 Students	1,000,000	< \$2.00 / mo	\$2,500.00	\$5,100.00	\$42,500.00	\$85,000.00	\$115,000.00
250,000 Students	2,500,000	< \$3.50 / mo	\$6,250.00	\$12,750.00	\$106,250.00	\$212,500.00	\$287,500.00
500,000 Students	5,000,000	< \$4.50 / mo	\$12,500.00	\$25,500.00	\$212,500.00	\$425,000.00	\$575,000.00
1,000,000 (1 Million)	10,000,000	< \$5.00 / mo (Max \$1,030 if 5% dynamic)	\$25,000.00 / mo	\$51,000.00 / mo	\$425,000.00 / mo	\$850,000.00 / mo	\$1,150,000.00 / mo

TABLE 2: TIDB CLOUD DATABASE MONTHLY COST SCALE (0 TO 1,000,000 STUDENTS)

Official TiDB Cloud Serverless Rates: 5 GiB Storage + 50M Request Units (RU) FREE each month. Overages: \$0.25/GiB storage, \$0.10 per 1M RU.

Scale (Active Students)	Estimated Storage (Books + History)	Monthly Request Units (RU)	TiDB Free Tier Coverage Status	Storage Overage Cost (\$)	Compute (RU) Cost (\$)	TOTAL TIDB MONTHLY COST (\$)	Equivalent in EGP (~50 EGP/\$)
0 – 1,000 (Launch)	0.4 GB	~250,000 RU	100% Free Quota	\$0.00	\$0.00	\$0.00 / mo	0 EGP
10,000 Students	0.8 GB	~1,800,000 RU	100% Free Quota	\$0.00	\$0.00	\$0.00 / mo	0 EGP
50,000 Students	2.5 GB	~9,000,000 RU	100% Free Quota	\$0.00	\$0.00	\$0.00 / mo	0 EGP
100,000 Students	4.8 GB	~18,500,000 RU	100% Free Quota (<5GB & <50M RU)	\$0.00	\$0.00	\$0.00 / mo	0 EGP
250,000 Students	10.5 GB	~45,000,000 RU	5 GB Free + 50M RU Free	\$1.38 (5.5 GB paid)	\$0.00 (under 50M RU)	~\$1.38 / mo	~69 EGP / mo
500,000 Students	21.0 GB	~92,000,000 RU	5 GB Free + 50M RU Free	\$4.00 (16 GB paid)	\$4.20 (42M RU paid)	~\$8.20 / mo	~410 EGP / mo
1,000,000 (1 Million)	38.5 GB	~185,000,000 RU	5 GB Free + 50M RU Free	\$8.38 (33.5 GB paid)	\$13.50 (135M RU paid)	~\$21.88 / mo	~1,094 EGP / mo

COMBINED INFRASTRUCTURE MONTHLY COST AT 1,000,000 STUDENTS: AI Engine (<\$5.00/mo) + TiDB Cloud Database (~\$21.88/mo) = ~\$26.88 / Month Total (~1,344 EGP / Month). This achieves a 99.89% cost reduction compared to unoptimized architectures.